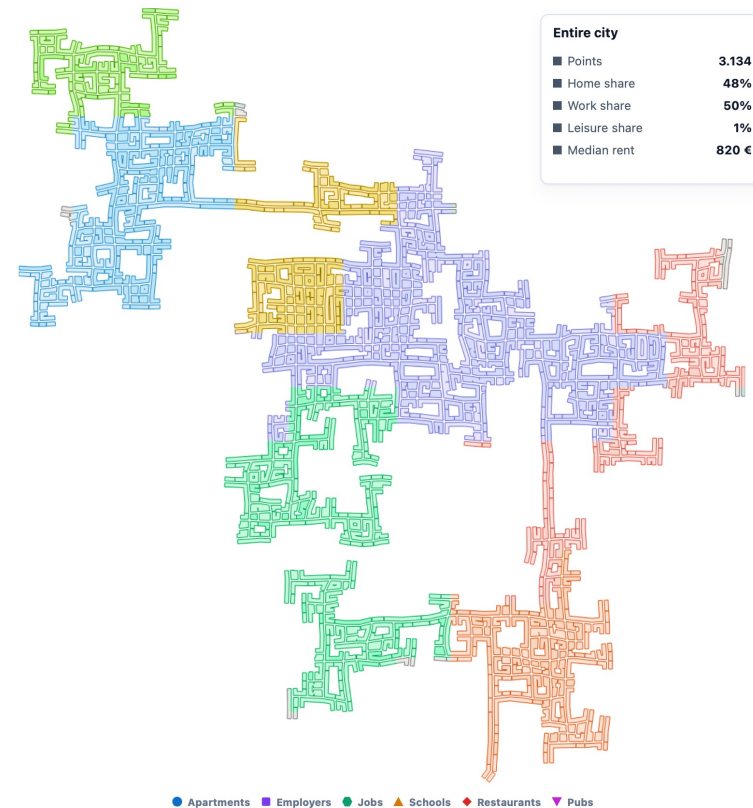


# High Performance Data Analytics & Visualization

## Challenge 2: Patterns of Life

Presented by Fabio Dollaku, João Cláudio Castro Ribeiro, Maurice Lesch, Kevin Moreira da Silva

CITY AREAS

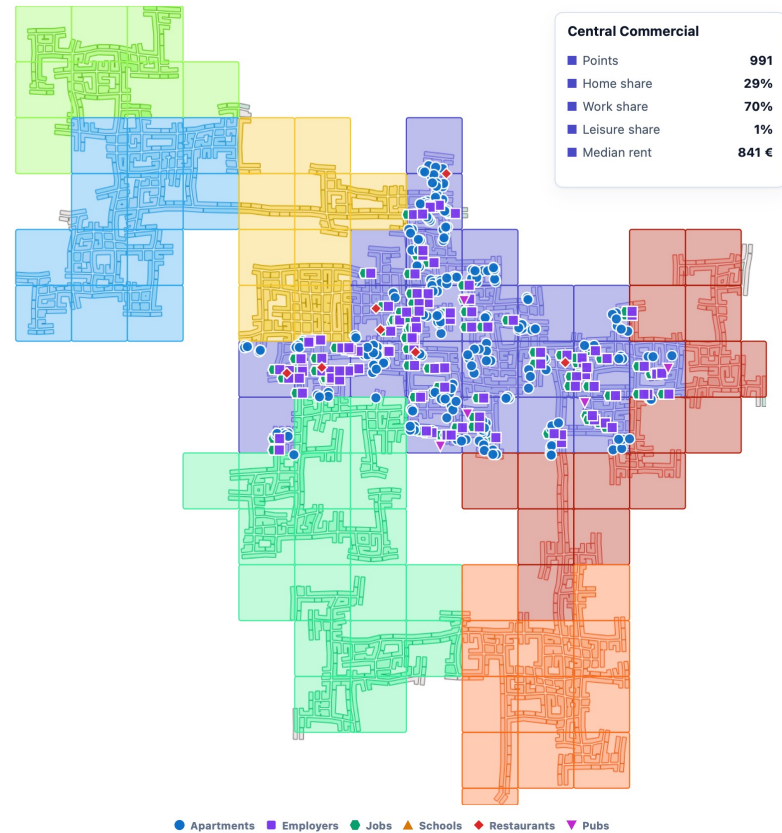


# **Characterization of distinct areas**

# Identified Urban Areas

- Shows 7 distinct city areas (based on 80x80 grids)
- Separates residential and commercial zones
- Preserves location, we can see where each area is
- Question requires spatial information
- Clustered areas are contiguous, not random groups
- Lets us connect the classification directly to the city layout
- Color-blind safe through different icon shapes

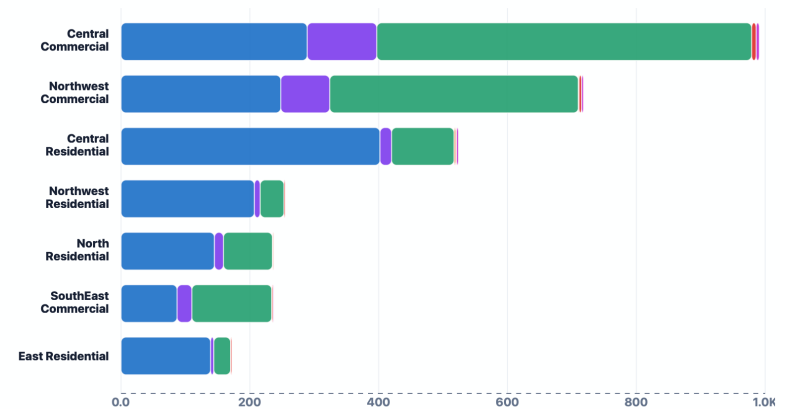
CITY AREAS



# Area Composition Analysis

- Counts apartments, employers, jobs, schools, restaurants, pubs
- Residential areas are apartment heavy
- Commercial areas are job/employer heavy
- Leisure activity is low everywhere
- Stacked horizontal bar chart
- Used because each area contains multiple feature types
- Stacked bars show both, total size and composition

WHAT IS IN EACH AREA?



## Entire city

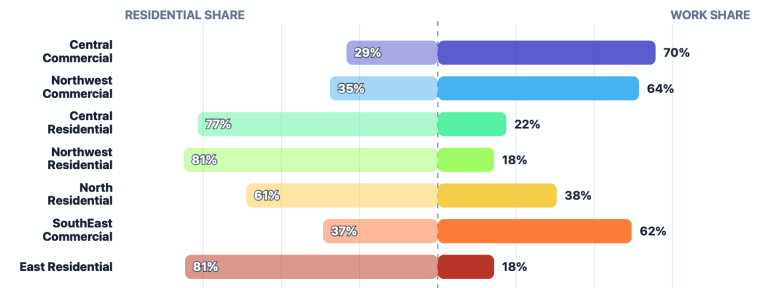
- Dominant use
- Active
- Apartments
- Employers
- Jobs
- Schools
- Restaurants
- Pubs

**Apartments (48%)**  
6 of 6  
1.517  
253  
1.328  
4  
20  
12

# Residential vs Work Distribution

- Directly compares home share vs work share
- Clearest representation to separate housing and working zones
- Used because the main contrast is home- vs work-oriented
- Diverging bars clearly visualize the imbalance between housing and the job market
- Changed opacity to show imbalance

## HOMES AND JOBS



### Entire city

|                     |          |
|---------------------|----------|
| Homes counted       | 1,517    |
| Work points counted | 1,581    |
| Work per apartment  | 1.0      |
| Balance gap         | +2.0 pts |
| ■ Apartments        | 1,517    |
| ■ Employers         | 253      |
| ■ Jobs              | 1,328    |
| ■ Schools           | 4        |
| ■ Restaurants       | 20       |
| ■ Pubs              | 12       |

# Socioeconomic Characteristics of Areas

- Compares activity, home share, work share, rent, and wage
- Highlights the strongest residential and commercial areas
- Supports the district labels with numeric evidence
- Shows many metrics at the same time
- Validates the area labels with several indicators
- Heatmap coloring makes strong and weak values easy to spot quickly
- Exact value lookup

## KEY AREA MEASURES

|                       | ACTIVITY | HOMES | WORK | LEISURE | RENT | WAGE |
|-----------------------|----------|-------|------|---------|------|------|
| Central Commercial    | 990      | 29%   | 70%  | 1%      | €841 | €14  |
| Northwest Commercial  | 720      | 35%   | 64%  | 1%      | €854 | €15  |
| Central Residential   | 520      | 77%   | 22%  | 1%      | €731 | €14  |
| Northwest Residential | 260      | 81%   | 13%  | 1%      | €772 | €18  |
| North Residential     | 240      | 61%   | 38%  | 1%      | €789 | €15  |
| SouthEast Commercial  | 240      | 37%   | 62%  | 1%      | €883 | €17  |
| East Residential      | 170      | 81%   | 13%  | 1%      | €822 | €17  |

### Entire city

|                  |       |
|------------------|-------|
| Activity         | 3.134 |
| Rent city median | €820  |
| Wage city median | €16   |
| Homes            | 48%   |
| Work             | 50%   |
| Apartments       | 1.517 |
| Employers        | 253   |
| Jobs             | 1.328 |

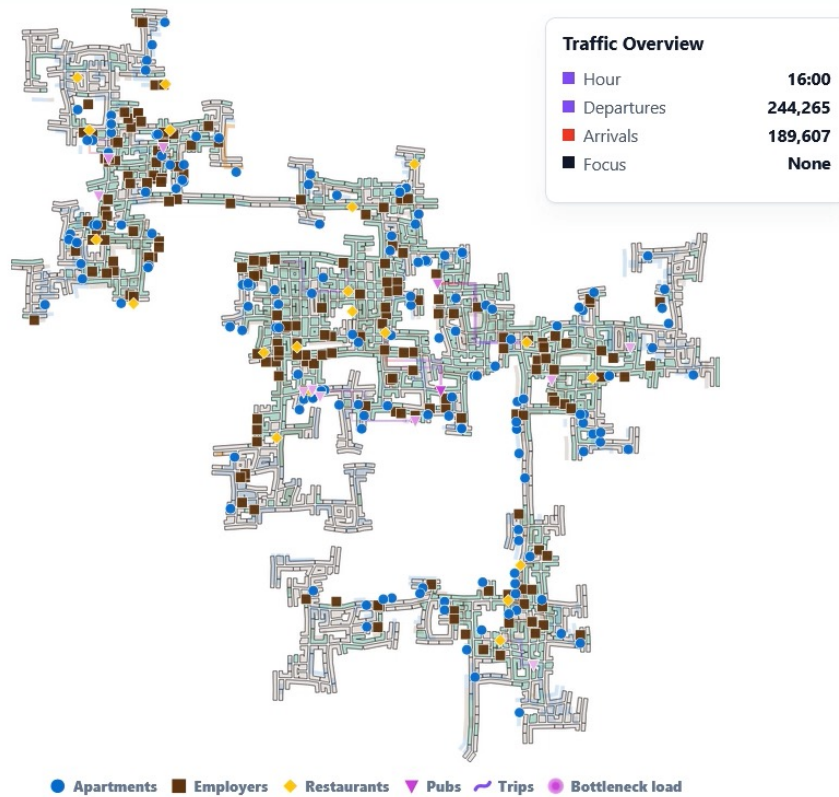
**Traffic & busiest  
areas in the city**

# Analysis overview

- Analysis on busiest Areas in Ohio & on traffic bottlenecks
- Relevant data: Check-in & Travel journals + Buildings & Pubs information
- Workflow:
  1. Departure / Arrival analysis
  2. Hotspot identification
  3. Pressure / Bottleneck identification

# Departure / Arrivals Analysis

CITY MOVEMENT MAP



HOURLY FLOW



Entire city

|                   |           |
|-------------------|-----------|
| Total trips       | 4,199,312 |
| Departures        | 2,099,656 |
| Arrivals          | 2,099,656 |
| Net flow          | 0         |
| Peak traffic hour | 16:00     |
| Peak departure    | 16:00     |
| Peak arrival      | 17:00     |



## Check-in journal + City mouvement map

Detect busiest areas  
Identify traffic patterns  
Categorization of the venues



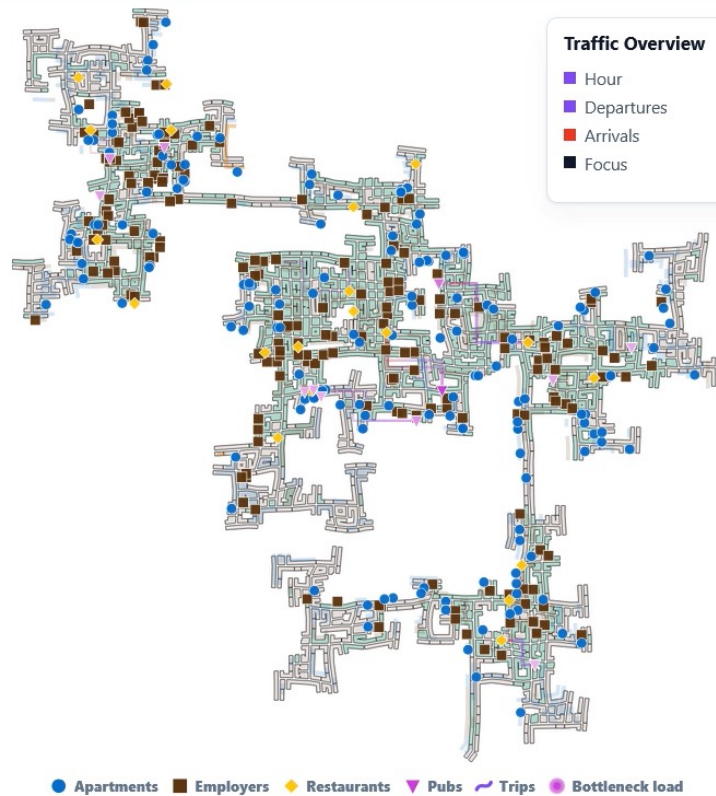
## Hourly flow

Trip & Activity analysis  
Matters for noise & interventions  
Semantic coloring + map markers

# Hotspot identification

Pressure ratio =  $\frac{\text{arrivals}}{\text{capacity}}$  | capacity: venue maxOccupancy

## CITY MOVEMENT MAP



## HOT TRAFFIC LISTS

### TOP HOTSPOTS (NUMBER OF CHECK-INS)

|                  |        |
|------------------|--------|
| Pub #1344        | 54,174 |
| Pub #1342        | 50,400 |
| Restaurant #1801 | 48,264 |

### ORIGIN-DESTINATION CORRIDORS (NUMBER OF TRIPS)

|                             |       |
|-----------------------------|-------|
| Pub #1342 → Restaurant #448 | 7,644 |
| Restaurant #448 → Pub #1342 | 7,264 |
| Pub #1344 → Restaurant #899 | 5,141 |

### DESTINATION PRESSURE (RATIO)

|           |        |
|-----------|--------|
| Pub #1344 | 902.87 |
| Pub #1342 | 654.53 |
| Pub #1800 | 510.11 |

# Hotspot identification



## Top hotspots

Number of checkins = Arrivals  
Matching color of arrivals  
List popular places



## Number of Trips

Rank OD corridor pairs  
Highlight departure & arrival  
Mainly depends on the number of trips



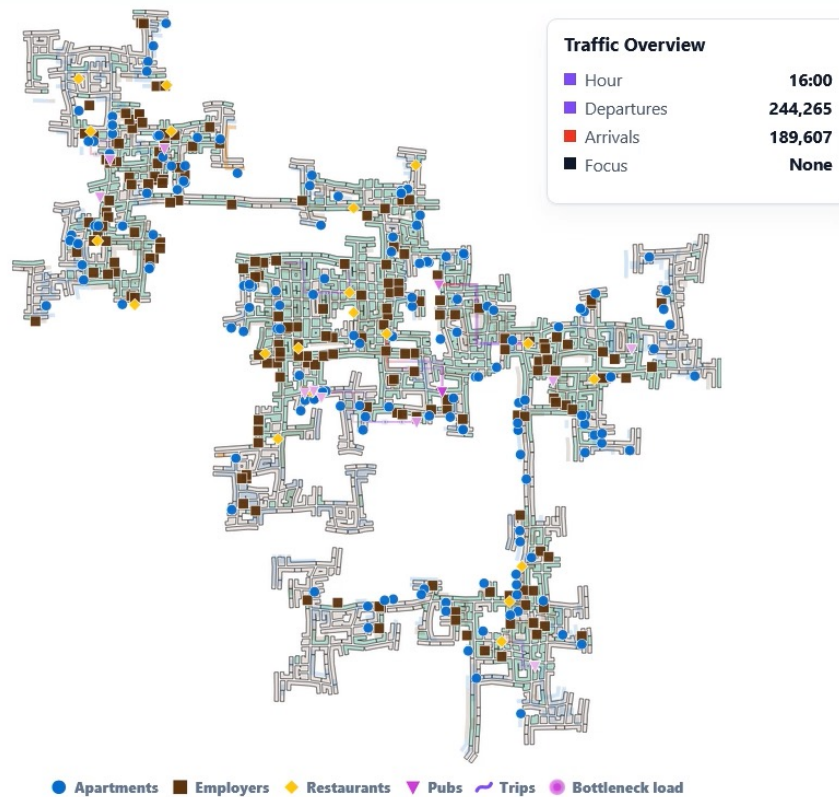
## Ratio

Identify location for high traffic bottleneck  
Reflect color of pubs  
Pubs = high activity place

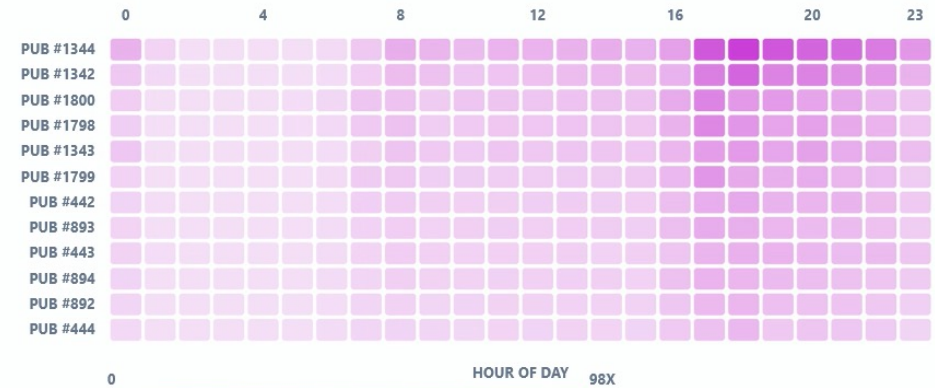
# Pressure / Bottleneck identification

$$\text{peak load} = \frac{\text{peak arrivals}}{\text{capacity}} \mid \text{total load} = \frac{\text{number of checkins}}{\text{capacity}}$$

## CITY MOVEMENT MAP



## HOURLY BOTTLENECK SEVERITY



## Location by hour

■ Worst location  
Peak pressure hour  
Peak load  
Peak arrivals  
Capacity

**Pub #1344**  
**18:00**  
**97.67x**  
**5,860**  
**60**

# Daily routines

# Compare daily routines

Compare participants:

#41 — age 41, HighSchoolOrCollege, joviality 0.86

▼

DAILY ROUTINE

#100 — age 29, Low, joviality 0.14

▼

#81 — age 27, Graduate, joviality 0.87

#82 — age 40, Graduate, joviality 0.96

#83 — age 53, HighSchoolOrCollege, joviality 0.03

Check-in count → Colour intensity

quantitative data → sequential colour scale



**More check-ins**

Venue type → colour hue

Nominal data → hue



**Apartment**



**Workplace**



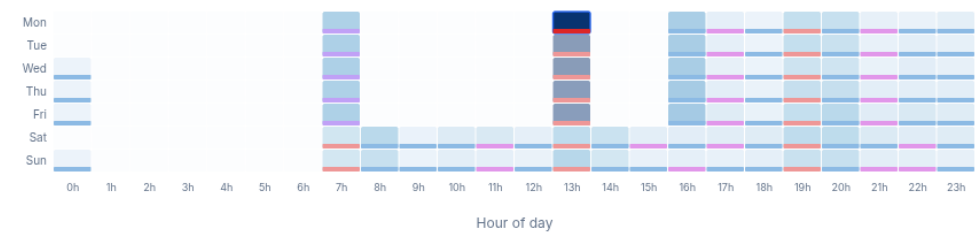
**Restaurant**



**Pub**

# Daily routine

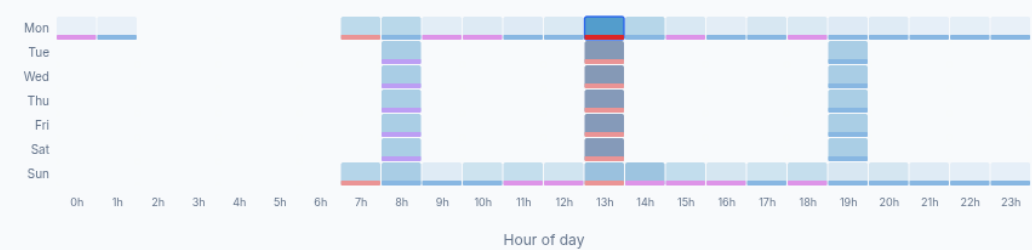
Participant #41



Monday 13:00

|           |               |             |            |
|-----------|---------------|-------------|------------|
| Source    | Daily Routine | Participant | P41        |
| Date      | None          | Weekday     | Monday     |
| Hour      | 13:00         | Venue       | Restaurant |
| Check-ins | 128           | Apartment   | 0          |
| Workplace | 64            | Restaurant  | 64         |
| Pub       | 0             |             |            |

Participant #100

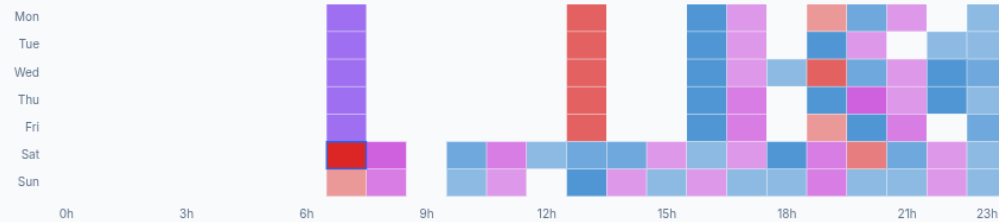


Monday 13:00

|           |               |             |            |
|-----------|---------------|-------------|------------|
| Source    | Daily Routine | Participant | P100       |
| Date      | None          | Weekday     | Monday     |
| Hour      | 13:00         | Venue       | Restaurant |
| Check-ins | 65            | Apartment   | 9          |
| Workplace | 0             | Restaurant  | 36         |
| Pub       | 20            |             |            |

# Monthly routine

Participant #41



Mar 2022 · Sat

|           |                 |             |          |
|-----------|-----------------|-------------|----------|
| Source    | Monthly Routine | Participant | P41      |
| Date      | 2022-03         | Weekday     | Saturday |
| Hour      | 7:00            | Venue       | 0        |
| Check-ins | 1               | Apartment   | 0        |
| Workplace | 0               | Restaurant  | 1        |
| Pub       | 0               |             |          |

Select desirable month

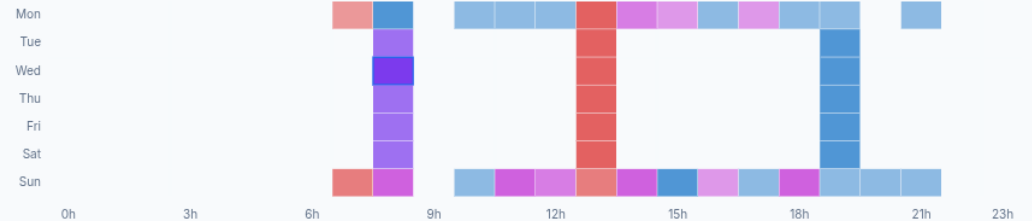
Month **Mar 2022** ▼

Mar 2022

Apr 2022

May 2022

Participant #100



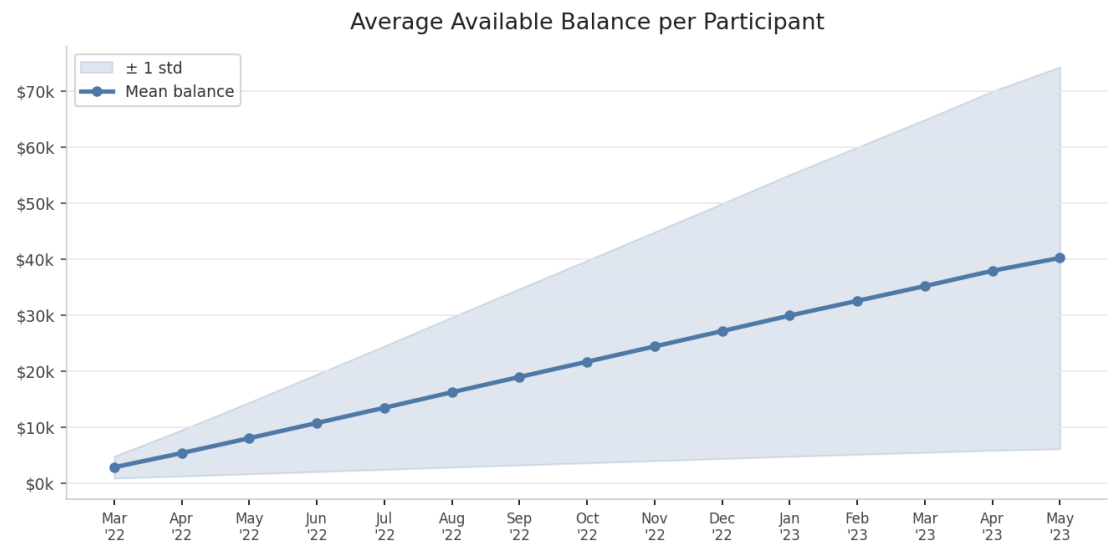
Mar 2022 · Wed

|           |                 |             |           |
|-----------|-----------------|-------------|-----------|
| Source    | Monthly Routine | Participant | P100      |
| Date      | 2022-03         | Weekday     | Wednesday |
| Hour      | 8:00            | Venue       | 0         |
| Check-ins | 5               | Apartment   | 0         |
| Workplace | 5               | Restaurant  | 0         |
| Pub       | 0               |             |           |

**Patterns over  
time in  
engagement**

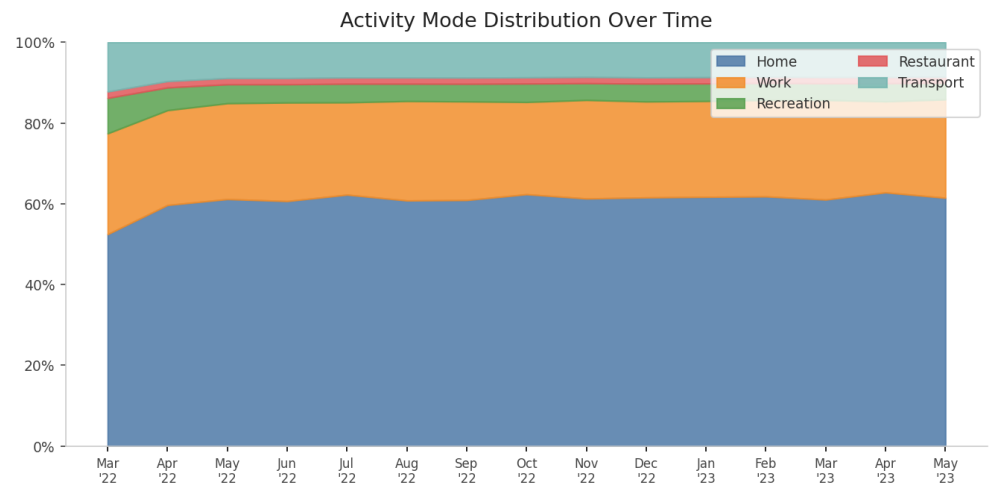
# Wealth Accumulation Over Time

- **Plot type: Line chart + uncertainty band**
  - Best for showing trends over time
  - Slope clearly highlights growth
  - Shaded band shows variation around the average
- **Why this works?**
  - Combines trend + spread in one chart
  - Easy to read over 15 months
- **Why not other charts?**
  - Bar charts hide trend flow
  - Box plots break continuity
  - Scatter plots add unnecessary noise



# How Residents Spend Their Time

- **Plot type: 100% Stacked area chart**
  - Shows changing activity proportions over time
  - Highlights part-to-whole relationships
  - Easy to compare category share across months
- **Why this works?**
  - Keeps focus on relative behaviour
  - Clear view of long term shifts
- **Why not other charts?**
  - Pie charts are hard to compare over time
  - Grouped bars become cluttered
  - Separate lines lose the 100% relationship



# Social Network Activity

- **Plot type: Dual-panel linked line chart**
  - Displays interaction volume and participation rate together
  - Shared timeline keeps trends connected
  - Separate panels preserve scale accuracy
- **Why this works?**
  - Avoids misleading comparisons
  - Maintains clarity between two metrics
- **Why not other charts?**
  - Dual y-axes can imply false correlation
  - Separate charts weaken comparison
  - Normalised charts hide actual volume

